AN IDEMPOTENT MODULO AN IDEAL IS LIFTED BY AN ETALE RING MAP THAT IS AN ISOMORPHISM MODULO THE IDEAL

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Etale ring map and its criterion through the module of differentials, here in the form that the derivative generates the unit ideal. used in the section, source lines 1740-1770, and applied at proof lines 1794-1797. Supplies the verification that A to A' is etale for the explicitly constructed A'.
- Idempotent of a ring as a root of t^2 - t. used throughout, at proof lines 1792-1803. Explains why adjoining a root of t^2 - t and inverting t - 1 + x is the right construction and why ebar is a root of the same polynomial modulo I.
- Quotient of a localisation by an ideal is the localisation of the quotient. used at proof lines 1798-1801 to compute A'/IA'. Identifies A'/IA' with A/I[t]/(t^2 - t) inverted at t - 1 + ebar, which is what makes the required isomorphism visible.

Source reference. Original lemma 07LY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9708–9708. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let A be a ring, let $I \subset A$ be an ideal, let $\overline{e} \in A/I$ be an idempotent. There exists an étale ring map $A \to A'$ which induces an isomorphism $A/I \to A'/IA'$ and an idempotent $e' \in A'$ lifting $\overline{e}$.*

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Proof. Choose any lift $x \in A$ of $\overline{e}$. Set

$$A' = A[t]/(t^2 - t)\left[\frac{1}{t - 1 + x}\right].$$

The ring map $A \to A'$ is étale because $(2t-1)\mathrm{d}t = 0$ and $(2t-1)(2t-1) = 1$ which is invertible. We have $A'/IA' = A/I[t]/(t^2-t)[\frac{1}{t-1+\overline{e}}] \cong A/I$ the last map sending t to $\overline{e}$ which works as $\overline{e}$ is a root of $t^2 - t$. This also shows that setting e' equal to the class of t in A' works. $\square$